The Prime-Gap Equivalence and the Integral-Shift Frontier

A Lean-checked research interface for Erdős Problem #251

Will Cook

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Abstract

Let $p_0 = 2, p_1 = 3, \ldots$ enumerate the primes and let $g_n = p_{n+1} - p_n$. We formalise in Lean both the exact finite identity

$$\sum_{i=0}^{n} \frac{p_i}{2^{i+1}} = 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}},$$

and, conditional only on summability, the infinite identity and irrationality equivalence

$$\sum_{i\geq 0} \frac{p_i}{2^{i+1}} = 2 + \sum_{i\geq 0} \frac{g_i}{2^{i+1}}, \qquad \mathrm{Irr}\left(\sum_{i\geq 0} \frac{p_i}{2^{i+1}}\right) \iff \mathrm{Irr}\left(\sum_{i\geq 0} \frac{g_i}{2^{i+1}}\right).$$

The prime number theorem supplies the summability in ordinary mathematics; that external input is not formalised here and remains an explicit hypothesis of the Lean declarations.

For an abstract dyadic tail recurrence $T_{N+1} = 2T_N - g_{N+1}$ with integer digits we prove the block identity $T_{N+h} = 2^h T_N - B_{h,N}$, and hence that the shift $T_{N+h} - T_N$ is integral *if and only if* $(2^h - 1)T_N$ is; that integrality of one shift propagates to every later index; and, by Euler's congruence, that a state with odd reduced denominator d has an integral shift of length $\varphi(d)$.

Problem #251 remains open. Its shortest surviving consumer is now exact: rationality would force some fixed $h \geq 1$ for which the actual tail shifts $T_{N+h} - T_N$ are integers at every sufficiently large N. Thus it is enough to prove cofinal non-integrality for every fixed h. We also prove a local certificate: two adjacent shifts lying strictly between -1 and 1 cannot both be integral when the corresponding prime gaps differ. Consequently it is enough to produce such adjacent small-mismatch pairs cofinally. The actual prime gaps are Lean-checked to be unbounded, hence not eventually periodic, but that fact alone supplies no tail smallness. The free-carry identity explains why generic periodicity or automaticity arguments cannot finish the proof; the missing theorem must control the full prime-gap tails.

Authorship and AI use. The prose in this version was generated by large-language-model agents under Will Cook's direction. Cook set the objectives and acceptance criteria, reviewed and authorised the manuscript, and is responsible for its contents. The agents are production tools, not authors. See *Statements and declarations*.

1 The problem

Let p_n denote the nth prime. Erdős Problem #251 asks whether

$$\Pi = \sum_{n \geq 1} \frac{p_n}{2^n}$$

is irrational [1][2, p. 62][3, p. 103]. Numbering and status follow Bloom's catalogue [4]. The problem is open.

Throughout the formal development the primes are indexed from zero, so that $p_0^{(0)} = 2$, $p_1^{(0)} = 3$, $p_2^{(0)} = 5$, and the gaps are $g_i = p_{i+1}^{(0)} - p_i^{(0)}$. In that indexing

$$\Pi = \sum_{i \geq 0} \frac{p_i^{(0)}}{2^{i+1}},$$

which is the convention every statement below uses; we drop the superscript from here on. A second normalisation with denominator 2^i also occurs, and at every finite horizon it is exactly twice this one,

$$\sum_{i=0}^{n-1} \frac{p_i}{2^i} = 2 \sum_{i=0}^{n-1} \frac{p_i}{2^{i+1}},$$

the factor-of-two normalisation. A factor of two does not change rationality, so no result below depends on the choice; the identity is stated so that the indexing cannot drift silently.

Relation to prior work. Erdős proved that $\sum_n p_n^k/n!$ is irrational for every $k \geq 1$ [1], and conjectured both that $\sum_n p_n^k/2^n$ is irrational for every k and that $\sum_{n\geq 1} p_n/(g_1 \cdots g_n)$ is irrational whenever $g_n \geq 2$ and $g_n = o(p_n)$ [3, p. 103]; the choice $g_n = p_n + 1$ shows that some growth condition is needed there. The statement of Problem #251 has been formalised before, as a conjecture with an unfilled proof, in the *formal-conjectures* collection. No claim of priority is made for anything below.

Two routes and where they stop. The digits p_n grow, so the series is not a digit expansion in any bounded alphabet, and the standard rationality criteria for such expansions do not apply directly. Two rewritings suggest themselves. The first replaces the primes by their gaps, which are bounded on average and are the natural object of prime-distribution theory; Section 2 carries this out exactly, and identifies the endpoint term that a passage to the limit must control. The second reads rationality as a constraint on a tail orbit; Section 3 develops that constraint, and Section 4 gives the exact reason it does not constrain the gaps themselves.

Status of everything below. *Status.* The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Statement	Status	Treatment here
Irrationality of Π	Open	Not proved.
Finite prime-gap identity	Proved here	Theorem 2.2, with the endpoint retained.
Infinite prime-gap identity	Lean-checked conditionally	Theorem 2.3; the prime number theorem supplies the hypothesis externally.
Prime-series/gap-series irrationality equivalence	Lean-checked conditionally	Corollary 2.4.
Block identity for the tail recurrence	Proved here	Theorem 3.2.
Integral shift criterion	Proved here; an equivalence	Theorem 3.3.
Totient shift from an odd denominator	Proved here	Theorem 3.4.
Propagation of an integral shift	Proved here	Theorem 3.5.
Eventual integral shift for every rational orbit	Lean-checked	Theorem 3.6.
Cofinal shift escape implies irrationality	Lean-checked abstractly	Theorem 3.7.
Actual prime gaps are unbounded and not eventually periodic	Lean-checked	Proposition 3.9.
Adjacent small-mismatch pair excludes simultaneous integrality	Lean-checked	Theorem 3.8.
Concrete prime-gap tail recurrence	Paper-proved from PNT	Not yet instantiated as a Lean infinite sum.
Rationality forces periodic gaps	False	Section 4; the carry is free.
Cofinal adjacent small-mismatch supply	Proposed sufficient theorem	Problem 5.2; not proved.

Reading map. Section 2 is the reformulation, Section 3 the tail algebra, Section 4 the no-go, and Section 5 the remaining obligation. Linked phrases open the corresponding Lean declaration at the pinned source revision 27ec97c4c2e2.

Keywords. irrationality; prime gaps; dyadic series; summation by parts; Lean 4. **MSC 2020.** 11J72 (primary); 11N05, 68V20 (secondary).

2 Summation by parts, with the endpoint retained

Write $\mathbb{N}_{>0}$-indexed sums with the convention

$$D(P,n) = \sum_{i=0}^{n-1} \frac{P(i)}{2^{i+1}}, \qquad \Delta(P,n) = \sum_{i=0}^{n-1} \frac{P(i+1) - P(i)}{2^{i+1}},$$

for a rational sequence P. These are the dyadic partial sum and the dyadic difference sum.

Proposition 2.1 (finite summation by parts). *For every rational sequence P and every $n \geq 0$,*

$$D(P, n+1) = P(0) + \Delta(P, n) - \frac{P(n)}{2^{n+1}}.$$

Proof. Induction on n. At $n = 0$ both sides equal $P(0)/2$. For the step, adding $P(n+1)/2^{n+2}$ to the left and $(P(n+1) - P(n))/2^{n+1}$ to the difference sum changes the endpoint term from $P(n)/2^{n+1}$ to $P(n+1)/2^{n+2}$, and the two adjustments agree. □

Formalised as the summation-by-parts identity. Nothing is assumed about P: no positivity, no monotonicity, and no convergence.

Specialising to $P(i) = p_i$, whose first value is $p_0 = 2$, and writing $g_i = p_{i+1} - p_i$ for the zero-based gaps, gives the reformulation.

Theorem 2.2 (prime-gap reformulation). *For every $n \geq 0$,*

$$\sum_{i=0}^{n} \frac{p_i}{2^{i+1}} = 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}}.$$

Formalised as the prime-gap summation by parts, using the gap partial sum and the gap cast identity; the latter records that the natural-number difference $p_{n+1} - p_n$ agrees with the difference taken in $\mathbb{Q}$, which needs $p_n \leq p_{n+1}$, the monotonicity step. The leading 2 is the first prime, not a normalising constant.

2.1 The infinite equivalence

Put

$$u_n = \frac{p_n}{2^{n+1}}, \qquad v_n = \frac{g_n}{2^{n+1}}.$$

The termwise identity $v_n = 2u_{n+1} - u_n$ is the dyadic discrete derivative. It makes the infinite passage a summable coboundary, rather than an informal deletion of the endpoint.

Theorem 2.3 (infinite prime-gap identity). *If $\sum_{n \geq 0} u_n$ is summable, then $\sum_{n \geq 0} v_n$ is summable and*

$$\sum_{n \geq 0} u_n = 2 + \sum_{n \geq 0} v_n.$$

Proof. The shifted sequence (u_{n+1}) is summable. Sum $v_n = 2u_{n+1} - u_n$ and use $u_0 = 1$:

$$\sum_{n \geq 0} v_n = 2\left(\sum_{n \geq 0} u_n - u_0\right) - \sum_{n \geq 0} u_n = \sum_{n \geq 0} u_n - 2.$$

□

The summability transfer is the gap-series summability theorem, and the displayed identity is the infinite prime-gap identity.

Corollary 2.4 (exact irrationality reformulation)**.** *Under the same summability hypothesis,*

$$\mathrm{Irr}\Big(\sum_{n\geq 0} u_n\Big) \iff \mathrm{Irr}\Big(\sum_{n\geq 0} v_n\Big).$$

The corresponding zero-based series with denominator 2^n is $4 + 2\sum_{n\geq 0} v_n$ and has the same irrationality status.

These are the normalised irrationality equivalence, the displayed-series identity, and the displayed-series irrationality equivalence.

Where the hypothesis comes from. The prime number theorem gives $p_n \sim n \log n$, hence $p_n/2^n$ is summable by comparison with any fixed exponential ρ^n with $1 < \rho < 2$. Consequently Theorem 2.3 and Corollary 2.4 are unconditional paper-level consequences of a classical external theorem. The local Lean development does not formalise the prime number theorem, so its declarations retain summability as an explicit hypothesis. This is an evidence boundary, not a remaining mathematical obstacle: the open content is exactly irrationality of the prime-gap series.

3 The tail recurrence and integral shifts

Rationality of a dyadic series constrains its tail orbit. This section proves what that constraint is, for an abstract recurrence.

Definition 3.1. Let $g : \mathbb{N} \to \mathbb{Z}$ and $T : \mathbb{N} \to \mathbb{Q}$. Say T satisfies the *dyadic tail recurrence* with digits g when

$$T_{N+1} = 2T_N - g_{N+1} \qquad \text{for every } N.$$

Write $\sigma_h(N) = T_{N+h} - T_N$ for the *shift* of length h at N, and call a rational number *integral* when it is the image of an integer.

These are the tail recurrence, the shift, and integrality. The digits are arbitrary integers. In the intended instance they are the prime gaps and T_N is the scaled tail of Π after level N, but that instantiation needs a summability argument and is not made here; every statement below is a theorem about Definition 3.1.

Iterating the recurrence accumulates an explicit integer. Define $B_{0,N} = 0$ and $B_{h+1,N} = 2B_{h,N} + g_{N+h+1}$, so that $B_{h,N} = g_{N+1}2^{h-1} + \cdots + g_{N+h}$: the tail block.

Theorem 3.2 (block identity)**.** *For every N and h,*

$$T_{N+h} = 2^h T_N - B_{h,N}, \qquad \textit{hence} \qquad \sigma_h(N) = (2^h - 1)\, T_N - B_{h,N}.$$

Proof. Induction on h; the step is one application of the recurrence together with $2 \cdot 2^h = 2^{h+1}$ and the recursion defining B. The second identity is the first minus T_N. □

Formalised as the iterated block identity and the scaled shift identity. The shift also obeys the recurrence in its own right, $\sigma_h(N+1) = 2\sigma_h(N) - (g_{N+h+1} - g_{N+1})$, the shift step identity.

Since $B_{h,N}$ is an integer, Theorem 3.2 converts a question about the shift into a question about a single scaled state.

Theorem 3.3 (integral-shift criterion)**.** *For every N and h, the shift $\sigma_h(N)$ is integral if and only if $(2^h - 1)T_N$ is integral.*

Proof. By Theorem 3.2 the two differ by the integer $B_{h,N}$, and subtracting an integer does not change integrality. □

Formalised as the integral-shift criterion, on the integer-shift invariance. This is an equivalence, not a one-way reduction: the block carries no information about integrality, so the two conditions are the same condition.

Theorem 3.4 (a shift of totient length)**.** *If the reduced denominator d of T_N is odd, then $\sigma_{\varphi(d)}(N)$ is integral.*

Proof. Since d is odd, 2 and d are coprime, so Euler's congruence gives $2^{\varphi(d)} \equiv 1 \pmod{d}$, that is $d \mid 2^{\varphi(d)} - 1$. Writing $2^{\varphi(d)} - 1 = dk$ and $T_N = u/d$ in lowest terms, $(2^{\varphi(d)} - 1)T_N = ku$ is an integer, and Theorem 3.3 transfers this to the shift. □

Formalised as the totient shift and the Euler multiplier. The hypothesis is a genuine restriction: the argument uses coprimality of 2 with the denominator, and the even part of a denominator is exactly what the doubling in the recurrence acts on.

Theorem 3.5 (propagation)**.** *If $\sigma_h(N)$ is integral, then $\sigma_h(N+k)$ is integral for every $k \geq 0$.*

Proof. By the shift step identity, $\sigma_h(N+1) = 2\sigma_h(N) - (g_{N+h+1} - g_{N+1})$ is an integer combination of an integer and two digits. Induct on k. □

Formalised as the one-step propagation and the propagation to every later index.

Theorem 3.6 (denominator collapse)**.** *Every rational-valued recurrence of Definition 3.1 has a fixed positive shift which is integral from some point onwards:*

$$\exists h \geq 1\ \exists N_0\ \forall N \geq N_0, \qquad \sigma_h(N) \in \mathbb{Z}.$$

Proof. Write the reduced denominator of T_0 as $2^s m$ with m odd. The block identity gives $T_s = 2^s T_0 - B_{s,0}$, so the reduced denominator of T_s divides m and is odd. Theorem 3.4 supplies the positive shift $h = \varphi(\operatorname{den} T_s)$ at s, and Theorem 3.5 keeps that shift integral forever. □

The factorisation and cancellation are the odd-denominator doubling lemma; the orbit form is the odd-denominator state theorem; and the quantified conclusion is the eventual integral-shift theorem.

The useful contrapositive is stated for a real recurrence. Call its shifts *cofinally non-integral* when, for every fixed $h \geq 1$ and every threshold N_0, some $N \geq N_0$ has $\sigma_h(N) \notin \mathbb{Z}$.

Theorem 3.7 (cofinal shift escape implies irrationality)**.** *Let $T : \mathbb{N} \to \mathbb{R}$ satisfy $T_{N+1} = 2T_N - g_{N+1}$ with integer g. If its shifts are cofinally non-integral, then T_0 is irrational.*

Proof. If T_0 were rational, the whole real orbit would be the cast of the rational orbit with that initial value. Theorem 3.6 would then produce one $h \geq 1$ whose shifts are integral at every sufficiently large index, contradicting cofinal non-integrality. □

Lean checks the real recurrence and shift interfaces as real dyadic recurrence and real shift, the rational-orbit transport as the rational-orbit cast theorem, cofinal non-integrality as the cofinal shift-escape predicate, and Theorem 3.7 as the irrationality consumer.

The actual prime-gap orbit. Define, at paper level,

$$\mathcal{T}_N = \sum_{j\geq 1} \frac{g_{N+j}}{2^j}.$$

The prime number theorem gives convergence, and an index shift gives

$$\mathcal{T}_{N+1} = 2\mathcal{T}_N - g_{N+1}.$$

If $G = \sum_{n\geq 0} g_n/2^{n+1}$, then $\mathcal{T}_0 = 2G - g_0 = 2G - 1$. Together with Theorem 2.3, this shows that Π, G, and $\mathcal{T}_0$ have the same rationality status. Thus Theorem 3.7 applies directly to Problem #251 once cofinal shift escape is proved for $\mathcal{T}$. This last analytic instantiation is proved here from the classical prime number theorem; it is not yet a Lean declaration. The abstract consumer, including its real/rational transport, is kernel-checked.

3.1 A local prime-gap certificate

The denominator-collapse theorem asks for cofinal non-integrality, but a specialist need not estimate the distance of a shift from every integer. Inside the open unit interval integrality collapses to equality with zero, and the one-step shift recurrence then exposes a single gap comparison.

Theorem 3.8 (adjacent small-shift obstruction)**.** *Let $T : \mathbb{N} \to \mathbb{Q}$ satisfy the dyadic tail recurrence with integer digits g. Fix h and N. If*

$$-1 < \sigma_h(N) < 1, \qquad -1 < \sigma_h(N+1) < 1, \qquad g_{N+h+1} \neq g_{N+1},$$

then $\sigma_h(N)$ and $\sigma_h(N+1)$ cannot both be integral. Consequently, if such a pair occurs beyond every threshold, the h-shift is not eventually integral.

Proof. An integral rational strictly between -1 and 1 is zero. If both shifts were integral, both would therefore vanish. Substitution in

$$\sigma_h(N+1) = 2\sigma_h(N) - (g_{N+h+1} - g_{N+1})$$

would give $g_{N+h+1} = g_{N+1}$, a contradiction. The cofinal statement chooses one such adjacent pair after the alleged onset of integrality. □

The finite contradiction is the adjacent small-shift obstruction; its quantified consumer is the cofinal small-mismatch theorem; and the actual-prime-gap specialisation is the prime-gap consumer. This is a local consumer, not a producer: no occurrence theorem for the two tail inequalities is claimed.

Proposition 3.9 (prime gaps do not become periodic)**.** *For every positive h, the actual consecutive-prime-gap sequence is not eventually periodic with period h.*

Proof. The factorial construction gives prime-free intervals of arbitrary length, so the consecutive-prime gaps are unbounded. An eventually periodic natural-valued sequence has finite range after its preperiod, while its finite initial segment is bounded as well; hence it is bounded, a contradiction. □

Lean checks the factorial argument as unboundedness of the actual gaps and the conclusion as non-eventual periodicity. Combining nonperiodicity with eventual strict smallness of one positive shift would also exclude eventual integrality, but eventual smallness at every sufficiently large index is stronger than the cofinal adjacent-pair supply in Theorem 3.8 and is not asserted here.

4 The carry is free

A natural hope is that rationality of a dyadic series forces its digit stream to be eventually periodic, or at least automatic, and that this is impossible for prime gaps. The hope fails at the first step, and the reason is a one-line identity.

Let $K : \mathbb{N} \to \mathbb{Q}$ be arbitrary and put $\kappa_n = 2K_n - K_{n+1}$: the carry coefficient.

Proposition 4.1 (exact telescoping)**.** *For every $n \geq 0$,*

$$\sum_{i=0}^{n-1} \frac{\kappa_i}{2^{i+1}} = K_0 - \frac{K_n}{2^n}.$$

Proof. Induction on n; the added term is $(2K_n - K_{n+1})/2^{n+1} = K_n/2^n - K_{n+1}/2^{n+1}$. □

Formalised as the carry telescoping identity, on the carry partial sum. When the carries are natural numbers and $K_{n+1} \leq 2K_n$, the coefficients are natural numbers and the two readings agree, the natural-carry cast.

The reading, and its exact status. The identity says that the carry sequence K is unconstrained: choose K freely, and the emitted coefficients κ_n have dyadic partial sums telescoping to K_0 up to $K_n 2^{-n}$. A rational limit therefore places no periodicity or automaticity requirement on the coefficient stream, because the same rational value arises from carry sequences that are not eventually periodic. This is why the tail constraint of Section 3 does not descend to the gap word: rationality controls a fractional orbit, and the integer part absorbs everything else.

Proposition 4.1 is proved; the accompanying claim that prescribed finite gap patterns can be realised by such a carry is a construction we have not formalised, and it is not used anywhere above. What is established is the identity and the consequence that the periodicity route is closed, not a classification of which coefficient streams occur.

5 Complements and further questions

Problem #251 is open. The exact surviving theorem is no longer an informal request for "randomness of the gaps".

Problem 5.1 (minimal prime-gap shift escape)**.** For every $h \geq 1$ and every N_0, prove that some $N \geq N_0$ satisfies

$$\sum_{j\geq 1} \frac{g_{N+h+j} - g_{N+j}}{2^j} \notin \mathbb{Z}. \tag{7.1}$$

The series in (7.1) is exactly $\mathcal{T}_{N+h} - \mathcal{T}_N$. Theorem 3.7 therefore turns any solution of Problem 5.1 directly into irrationality of Π. The quantifier over h is essential: a hypothetical rational value chooses a shift length from the odd part of its reduced denominator, and that length is not known in advance.

Problem 5.2 (cofinal adjacent small mismatch)**.** For every fixed $h \geq 1$ and every N_0, prove that some $N \geq N_0$ satisfies

$$\begin{aligned} \left|\sum_{j\geq 1} \frac{g_{N+h+j} - g_{N+j}}{2^j}\right| &< 1, \\ \left|\sum_{j\geq 1} \frac{g_{N+h+1+j} - g_{N+1+j}}{2^j}\right| &< 1, \\ g_{N+h+1} &\neq g_{N+1}. \end{aligned} \tag{7.2}$$

The two sums are adjacent h-shifts. Theorem 3.8 turns each instance of (7.2) into a finite contradiction to simultaneous integrality. A cofinal supply therefore proves Problem 5.1. This formulation deliberately asks only for sporadic adjacent pairs; the stronger assertion that a fixed shift is eventually always smaller than one is unnecessary.

A finite-block route to the inequalities. For $L \geq 1$ put

$$S_{h,N,L} = \sum_{j=1}^{L} \frac{g_{N+h+j} - g_{N+j}}{2^j}.$$

If $M(n) \geq g_n$, the discarded tail has absolute value at most

$$R_{h,N,L}(M) = \sum_{j>L} \frac{M(N+h+j) + M(N+j)}{2^j}.$$

Hence it is enough, cofinally in N for every fixed h, to find an $L = L(N,h)$ such that

$$\operatorname{dist}(S_{h,N,L}, \mathbb{Z}) > R_{h,N,L}(M). \tag{7.3}$$

Taking the classical bound $M(n) \ll n \log n$, a choice $L = A \log_2(N+h+2)$ with any fixed $A > 1$ makes the right side a negative power of N up to logarithms. Thus (7.3) asks for a finite dyadic anti-concentration estimate on a logarithmic-length block, not control of an infinite tail and not eventual periodicity of the full gap word. For Problem 5.2, the same truncation must certify two adjacent full-tail values inside the open unit interval, together with the displayed gap mismatch. A finite prefix is useful only when its omitted tail is rigorously dominated.

Candidate mechanisms and disconfirmers. A useful prime-distribution input would control the weighted difference word

$$(g_{N+h+1} - g_{N+1}, \cdots, g_{N+h+L} - g_{N+L})$$

modulo powers of two, or would force two completions of a long admissible prime pattern whose values of $S_{h,N,L}$ occupy separated dyadic cells. Isolated small gaps, isolated large gaps, average gap estimates, and the occurrence of any one fixed finite pattern do not suffice: both inequalities in (7.2) contain the complete infinite continuation. Nor does parity: after the first gap all g_n are even. Unboundedness and non-eventual-periodicity of the actual gaps, though now checked, do not imply the required small-tail recurrence. Section 4 is the decisive disconfirmer for any argument that tries to deduce automaticity or periodicity from the rational fractional orbit alone.

Specialist handoff. *Wanted:* for each fixed $h \geq 1$, prove the cofinal adjacent small-mismatch statement (7.2); independently, formalise the actual prime-gap infinite tail and its recurrence from summability. *Already checked:* the infinite prime/gap equivalence, removal of every power-of-two denominator, eventual propagation of one integral shift, and the real-orbit cofinal-escape consumer; actual prime gaps are unbounded and not eventually periodic; and one adjacent small-mismatch pair excludes simultaneous integrality exactly. *Most useful answer:* a theorem supplying cofinally many pairs in (7.2), with complete hypotheses and an effective or qualitative tail bound. A periodicity theorem, an automaticity theorem, or the occurrence of one isolated finite gap word is not enough. *If answered:* the checked local consumer contradicts the eventual integral shift forced by rationality; Theorem 3.7 then gives irrationality of the gap series, Corollary 2.4 transfers it to Π, and Problem #251 is solved once the actual-tail recurrence bridge is formalised.

Remaining formal boundary. The abstract irrationality consumer and the prime-specific local small-mismatch consumer are Lean-checked. The local source still takes summability of the prime series as a hypothesis and does not define $\mathcal{T}_N$ by an infinite sum; the prime-number-theorem comparison and the identification of that concrete tail with the checked real recurrence are paper proofs. No theorem supplies the cofinal pairs in (7.2), and none of the proposed prime-distribution mechanisms above is claimed or formalised.

Statements and declarations

Artefact and data availability. The pinned formal-source revision contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. This manuscript provides navigation rather than proof authority.

Declaration of generative AI use. Large-language-model agents were used throughout development to draft and revise prose, formal proofs, and software. The author set the objectives and acceptance criteria, selected and reviewed the claims, and approved the published version. The author assumes responsibility for the accuracy, interpretation, and presentation of the work. Generative systems are production tools; they are not authors and supply no independent authority. Lean checks each proof term against the fixed library version, and the sources linked here contain no proof placeholders and no project-defined axioms; Lean does not authorise the exposition, the citation choices, or the interpretation, for which the author remains responsible.

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A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision 27ec97c4c2e2. All declarations of this note live in one module. The summation-by-parts declarations are prime-specific; most of Section 3 is stated for arbitrary integer digits and arbitrary rational or real orbits, while Section 3.1 records the actual-gap specialisation. The concrete prime-gap tail and its convergence are paper-level consequences of the prime number theorem; they are not definitions in the pinned Lean module.

References

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[4] T. F. Bloom, *Erdős Problem #251*, `erdosproblems.com/251`, accessed 22 July 2026.